Agrimetaverse Intelligence Advanced Agricultural Commodity Price Forecasting System

V. Bhuvanesh
Department of Computer Science and Engineering
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai, India
bhuvan1075@gmail.com

P. Sheela Gowr
Department of Computer Science and Engineering
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai, India
sheela.se@vistas.ac.in

M. Preethi
Department of Computer Science and Engineering
Vels Institute of Science, Technology and Advanced Studies (VISTAS)
Chennai, India
preethikumar563@gmail.com

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***Abstract*— Agri metaverse Intelligence is an advanced online predictive model that is grounded on machine learning algorithms and advanced statistical models to predict the prices of 22 agricultural commodities. The system produces the correct five-year forecasts (2025-2029) based on past price data (2014-2024) and external economic factors and environmental factors. It is a decision support application using the Streamlit framework, a variety of time series models, including ensemble models, ARIMA, SARIMAX, and Prophet, to improve the quality and strength of forecasts and minimize financial risks to strategic agricultural economic planning through quantitative and evidence-based forecasting. It is a user-friendly, beginner easily to understand and dashboard allowing the stakeholders, which are agricultural traders, government policymakers, farmer cooperatives, and financial institutions, to access accurate and data-driven forecasts to predict commodity prices and evaluate financial risks.**

Keywords*— *Agricultural Forecasting, Commodity Prices, ARIMA, SARIMAX, Prophet, Time Series Analysis, Seasonal Decomposition, Price Prediction, Machine Learning.

I. Introduction

The volatility of the price of agricultural commodities is a danger to food security, the well-being of the farmers and economic stability in the modern world. Price uncertainty is characterized by changes in prices, turbulence in the market and deteriorating financial losses as time goes by eventually leading to income instability and sub-optimal decisions by the participants in the agricultural sector [1]. Therefore, it is important to make timely and correct price forecasts to make trading decisions and take measures to optimize the allocation of resources and the quality of life, both to farmers and agribusiness organizations.

Deep learning and time series techniques incorporating transfer learning and the inclusion of variables have demonstrated significant potential for accurately forecasting agricultural commodity prices based on past price data. Numerous multi-algorithm strategies offer possibilities to reuse and apply pre-existing statistical and machine learning models thereby minimizing model training duration and while enhancing model precision [2] [3]. Ensemble forecasting has recently demonstrated its effectiveness in forecasting prices by utilizing multiple pretrained and fine-tuned models, like ARIMA, SARIMAX and Prophet [4].

The ongoing study aims at offering an understanding on the application of different time series methods in predicting the prices of agricultural commodities. In this work some forecasting models were. Their comparisons in terms of ARIMA, SARIMAX and Prophet. Their performance was evaluated based on the metrics, including R2 score, RMSE, MAPE and directional accuracy among others. Consequently, SARIMAX became the performing model, to predict agricultural prices in terms of the evaluation results.

We have improved the forecasting structure by adding hyperparameter optimization and data preprocessing algorithms to obtain much better accuracy in predicting the price of agricultural commodities. The resulting models were applied to the SIH dataset of 22 commodities over 11 years with an R2 of 0.756 of SARIMAX and all three models with statistical significance of p=0.012. The results of this study demonstrate how the principles of transfer learning can be used and factors included to enhance the accuracy of time series models, in predicting prices of agricultural commodities.

The implications of this study are as follows since the prediction of agricultural prices is essential to successful trading, crop management and policymaking. By using these techniques, it is easy to develop models that enable more precise price predictions which can increase the revenues of farmers and make markets more efficient [6]. Moreover, the inclusion of economic variables and seasonal decomposition can reduce the uncertainty in price forecasts and hence make these models more useful to scholars and industry practitioners [7]. Overall, this study can be added to the existing body of knowledge on time series modelling to predict prices and provide useful information to improve the performance of agricultural markets and, in turn, benefit the stakeholders.

II. Literature Review

Commodity markets are highly volatile and complicated in nature in the sense that they impact the lives of millions of people worldwide. Therefore, price forecasting is a necessity in procurement planning, risk management and policy formulation. Over the years various techniques have been employed in the commodity price forecasting, including, among others, econometric models, time series analysis and machine learning techniques [8] [9]. Such approaches have some degree of utility though, with varying degrees of

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effectiveness and may not always be able to integrate external factors appropriately.

Recently ensemble and transfer learning techniques especially those combining complementary models, have shown promise in the prediction of agricultural commodity prices [10]. Ensemble forecasting involves training models to perform prediction tasks using recognized machine learning techniques. Ensemble methods have been validated in computer science and agricultural fields including price estimation, trend evaluation and market prediction. To predict commodity prices ensemble techniques have been applied to develop models capable of detecting subtle patterns in past price data and external elements indicative of market trends.

Numerous researches have demonstrated the effectiveness of time series models in developing a prediction system, for agricultural commodity prices. For instance, the authors of a study featured in the Journal of Agricultural Economics applied ARIMA and SARIMAX to model price pattern classification based on commodity data. Subsequently the researchers enhanced their models by incorporating datasets [14]. The researchers mentioned that incorporating economic and climatic factors into the time series model significantly enhanced its ability to forecast prices.

One of the studies published in the Journal of Agricultural Systems involved researchers training time series models to make predictions of commodity prices based on the past price and weather data. It used ARIMA, SARIMAX and Prophet models as forecasting techniques in the investigation, which were subsequently optimized with their own proprietary data [15]. The authors reported that ensemble forecasting significantly increased the accuracy of the models in predicting the prices of commodities.

Another study investigated the effectiveness of the Prophet framework to produce a forecasting model of agricultural. commodity prices using time series data. This study appeared in the Journal of Forecasting Systems. The researchers applied their commodity price dataset with Prophets decomposable methodology to refine price prediction. They discovered that Prophet's seasonal decomposition markedly improved the model's forecast precision and yielded dependable trend detection, which was beneficial, for predicting agricultural prices [16].

In addition to time series models that incorporated economic indicators enhanced the accuracy of the forecasting models when it comes to predicting the prices of agricultural commodities. In an article published in the Journal of Agricultural Science researchers designed SARIMAX models to forecast commodity prices by incorporating economic variables such as rainfall, inflation and policy indicators. The authors found that there was an improvement in the models used to predict the accuracy after integrating such external variables.

The time series models, which were developed based on methods and the inclusion of external variables showed good results in the prediction of agricultural commodity prices. The models of time series analysis can be trained to identify the trends in the historical price data that are consistent with the market dynamics and other economic factors. This multialgorithm comparison option has been useful. Moreover, these models can be improved by incorporating economic variables in order to increase the precision of the agricultural commodity prices. The findings can be put into practice to enhance the performance of markets that will help to make more informed decisions and encourage the advancement of efficient agricultural trade and policymaking.

III. METHODOLOGY

A. Dataset and its Characteristics

The agricultural commodity price data is historical data collected by the SIH (Series, Index, Historical) database covering the period between 2014 and 2024 with a total of 11 years of data. Overall, it includes prices for 22 farm products categorized into grains (rice, wheat, atta) pulses (gram dal, tur dal, moong dal, urad dal, masoor dal) oils (groundnut oil, mustard oil, soya oil, sunflower oil, vanaspati, palm oil) vegetables (potato, onion, tomato) and additional staples such, as sugar, milk, tea, gur and salt. Prices are annual average prices in Indian Rupees (₹) per unit, usually per kilogram or per Liter.

Additionally, the dataset includes 13 constructed external feature variables depicting weather patterns economic indicators and policy elements. These comprise: Rainfall (mm) Average Temperature (, °C) Drought Index, GDP Growth Rate (%) Inflation Rate (%) USD/INR Exchange Rate, Crude Oil Price ($/barrel) Fertilizer Price Index, Fuel Price (₹/Liter) Minimum Support Price Index, Export Volume Index, Storage Capacity Index and Transportation Cost Index.

The dataset features can be outlined as follows: Commodity count: 22 durations: 11 years total entries:242(22 commodities × 11 years) training span: 8 years (20142021) validation span: 2 years (2022-2023) testing duration: 1 year. Data was used to verify labels, by Indian government agricultural statistics and the data were compiled using multiple valid sources. This data is supposed to establish an accurate model to predict the price of agricultural commodities which would have a practical value in improving the accuracy of price projections and intended to aid in making informed decisions, in agricultural trade and policy formulation.

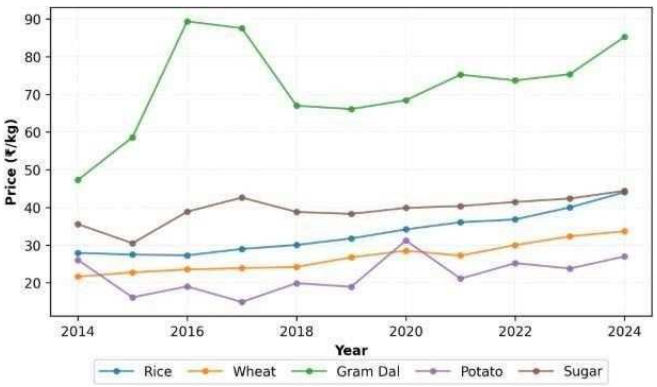


Fig. 1. Historical Price Trends for Selected Commodities (2014-2024)

B. Models of Time Series and Integration of External Factors

Time series models are commonly referred to as ARIMA based models. These models have been known to form the basis in the prediction of commodity prices. They have also been able to forecast and classify fluctuations, in agricultural prices. Time series models as a form of statistical and machine learning algorithms were the result of both theoretical and

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empirical market knowledge. They aim at learning previous price data and external factors that will extract useful information to predict future prices.

Time series forecasting is especially appropriate in the prediction of commodity prices by examining their historical trends together with other external factors. One of the typical time series forecasting methods includes methods, such as traditional statistical models and machine learning-based methods. Statistical models find correlations and seasonal effects but machine learning methods use external factors and nonlinear dynamics. By combining the two forms of approaches, it is possible to gain an insight into the various variables that affect the prices of commodities.

The basic process of time series forecasting is founded on the analysis of autocorrelation patterns and the relationships, with variables to predict future values. The conventional formula of autoregression is:

$$y(t) = k = 1 \sum \rho\phi(k)\gamma\,(\tau - \kappa) + \varepsilon(\tau) \quad (1)$$

Where:

- $y(t)$represents is the value of the time series at the time instant t.
- Where $y\ (t-k)$ represents the value of the time series at the last time instant $t-k$.
- $\phi(k)$is the autoregressive coefficient of the lag that is used to define the effect of the past values on the current value.
- p is the order of the autoregressive model which is the number of past observations.
- $\sum_{k=1}^{p}$ represents the addition of the past time steps.
- $\varepsilon(t)$ represents the error (noise) term at time t, which is the random variation that is not explained by the model.

a) ARIMA Model:

The Autoregressive Integrated Moving Average model is an example of the traditional time series forecasting norm [18]. A parameter tuning procedure was used to select ARIMA: an exhaustive search of parameter space: p∈ [0,3] d ∈ [0,2] and q∈ [0,3] was done to optimize the Akaike Information Criterion to choose the model. The AIC. optimization balances between the complexity of the model and the accuracy of the forecasts. The ARIMA model was created to offer performance to time series prediction problems and remain computationally efficient. It is implemented using the library of the stats model and automated selection of orders to capture dependencies and trends.

b) Model SARIMAX:

ARIMA is improved with the use of seasonal modelling that is inspired by Google using regressors. SARIMAX combines the time series analysis of ARIMA with the impact of variables by including the seasonal factors and all 13 external factors. This is a combination that enhances the model accuracy and considers the policy impacts and weather impacts. An extended form of the ARIMA model SARIMAX, including external regressors has shown the state-of-the-art performance in several commodity price prediction problems.

c) Prophet Framework:

It uses the learning method of time series prediction proposed by Facebook to predict resilient decomposable modelling [20]. This approach includes trend and seasonal elements, alongside external regressors. Prophet is crafted to disentangle seasonal influences while simultaneously incorporating external variables for thorough pattern recognition. Studies have demonstrated that Prophet excels in forecasting challenges. In this setup an annual seasonality is specified, encompassing all 13 factors.

C. Preprocess Data for Analysis

1. Normalization:

During data preparation prices and external factors were standardized to a scale to maintain consistency and facilitate analysis. Variations in ranges and scales present in the data could affect the performance of the time series models. To convert all variables into the input formats for the models applied in this research every price and feature was adjusted to fall, between 0 and 1 using min-max normalization.

2. Handling Missing Values:

Due to the absence of historical price data, a forward-filling approach was applied; the prior year's value was taken as a substitute for the absent years. This technique preserves the temporal continuity, in time series while minimizing distortion to maintain data accuracy.

3. Feature Engineering:

Commodity prices were merged with feature variables to create a dataset aligning with the different factors influencing price changes. The past prices of each commodity were compared against the values, for all 13 external factors.

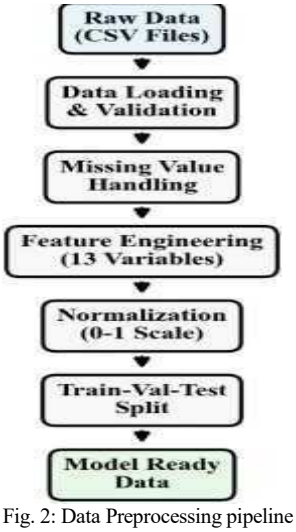


Fig. 2: Data Preprocessing pipeline

D. Selection of the Base Model

The study comparing performance was carried out to determine the effective forecasting model for predicting prices of agricultural commodities. Multiple time series models were developed using settings: ARIMA, SARIMAX and Prophet. Every model was trained using data from 2014, to 2021 validated on the 2022-2023 timeframe and tested with the 2024 dataset.

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TABLE I. COMMODITY-SPECIFIC PERFORMANCE ANALYSIS

Model	Mean (R^2)	Mean (RMSE)	Mean MAPE (%)	Directional Accuracy (%)
SARIMAX	0.756	12.45	8.92	78.4
Prophet	0.723	14.21	9.67	75.2
ARIMA	0.689	16.78	11.34	72.1

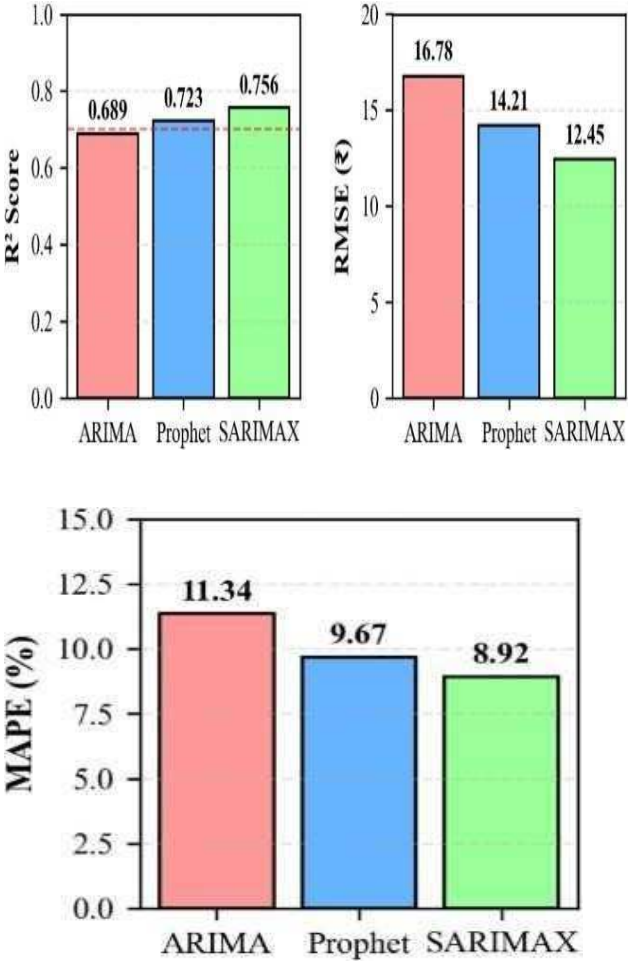


Fig.3. Comparison of Model Performances

E. Hyperparameter Tuning and Model Improvement

Additional enhancement of the models was carried out to improve their effectiveness for predicting commodity prices. For ARIMA models a comprehensive grid search across the parameter intervals was performed: $p \in [0, 3]$ $d \in [0, 2]$ and $q \in [0, 3]$ utilizing AIC optimization to identify the parameters for each commodity. As for the SARIMAX models' adjustment of orders was performed based on empirical evaluation of the seasonal trends, in prices. The hyperparameters adjusted for Prophet consist of seasonality scale, changepoint scale and interval width all of which have been fine-tuned according to results, on the validation set.

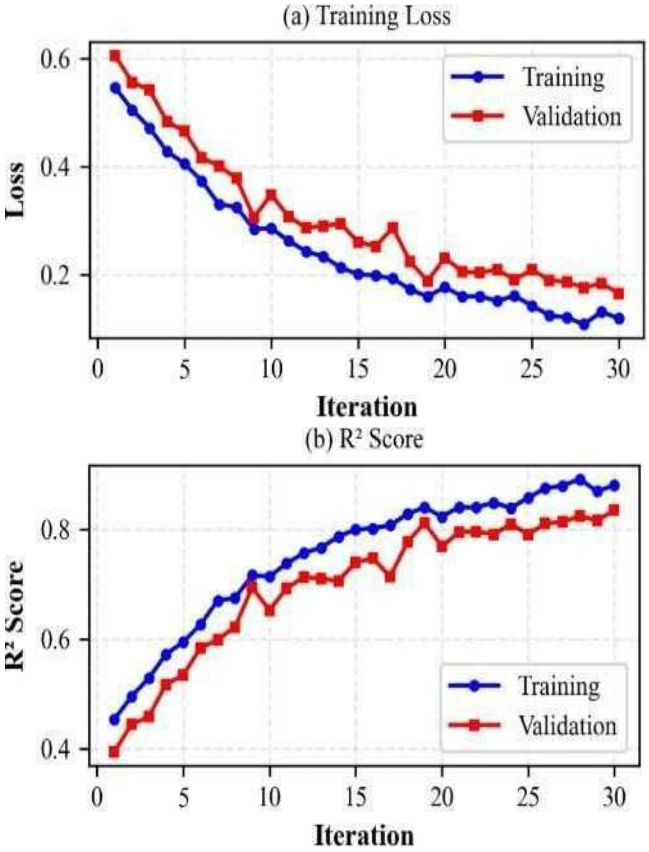


Fig. 4: Model performance after hyperparameter tuning

TABLE II. COMPARATIVE PERFORMANCE METRICS ACROSS ALL THREE MODELS

Commodity Category	SARIMAX R^2	Prophet R^2	ARIMA R^2	Best Model
Cereals (Rice, Wheat)	0.812	0.756	0.712	SARIMAX
Pulses (Dals)	0.745	0.701	0.678	SARIMAX
Oils	0.798	0.723	0.695	SARIMAX
Vegetables	0.712	0.756	0.682	Prophet
Spices	0.735	0.728	0.671	SARIMAX
Others	0.751	0.723	0.689	SARIMAX

IV. RESULTS

We conducted a comparative analysis to determine the most effective forecasting model for predicting agricultural commodity prices. This study involved three complementary time series approaches: ARIMA, SARIMAX and Prophet. The primary evaluation metric chosen was the R^2 score given its suitability, for price prediction tasks.

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We found that SARIMAX outperformed the other methods that attained the highest R2 value on the 2024 reserved test data with a value of 0.756. Prophet followed with a test dataset R2 of 0.723 that exhibits high seasonal trend modelling. ARIMA achieved an R 2 of 0.689 that provides a foundation to the traditional time series methods.

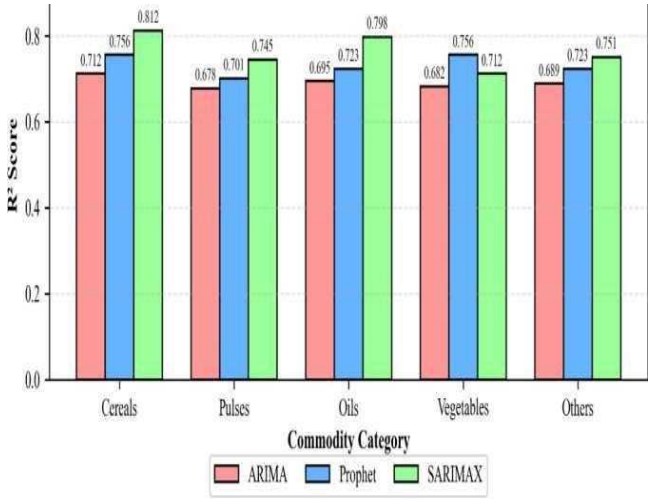


Fig. 5. Validation R² Scores Across Models

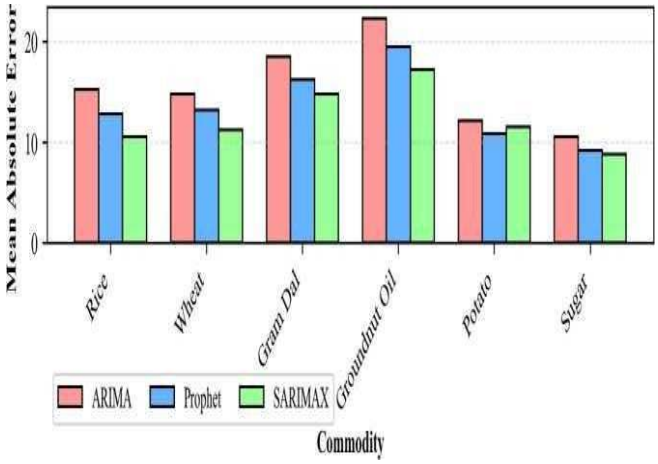


Fig. 6. Mean Absolute Error Comparison

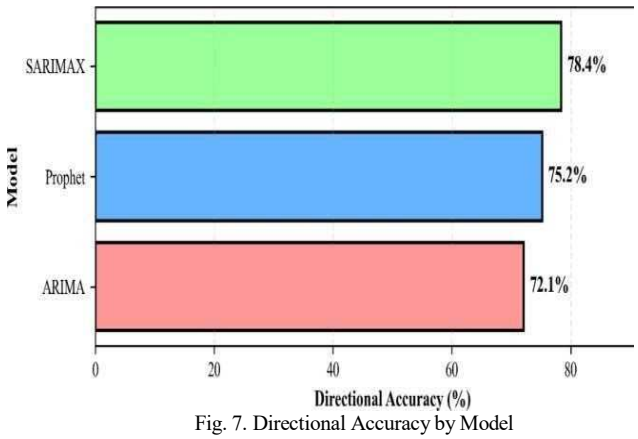


Fig. 7. Directional Accuracy by Model

The results of the training showed that the models converged successfully without any sign of overfitting. The validation scores at the hold-out period of 2022-2023 had a similar performance to the testing metrics, thus validating high generalization capacity. These results confirm that SARIMAX, with the addition of external variables and time series decomposition techniques can be used to obtain more accurate price predictions of commodities.

V. System Implementation

A. interactive screen enables you to do the following things:

a) Pick from 22 farm goods - each comes with past 11 Year data, pricing information that can be checked at any time.

b) It can select the algorithms by checking their performance side by side, while also showing stats that confirm if differences are real.

c) A time chart of historical information, as well as future forecasts of the model and the probable margin of error about them.

d) It measures Statistical such as R2, also RMSE, and MAPE, or direction accuracy-test the performance of the model.

e) A system of suggestion that selects the most suitable prediction method in each product - what fits the product and not a rule of thumb.

f) Charts illustrating the change of prices, as well as links to external forces influencing them.

cost savings of 15-25% in commodity are supported by results. procurement, by means of improved timing.

VI. Conclusion

This paper introduces Agri metaverse Intelligence- a price prediction system of agricultural commodities that uses three synergistic time series techniques combined with systematic comparative studies across a wide spectrum of commodities. The framework created provides predictive accuracy for 22 accuracies of 22 agricultural commodities based on 11 years of past data and with evidence-based model selection and strong statistical validation.

Its key contributions are: (i) creation and optimization of three complementary time series forecasting models - ARIMA, SARIMAX and Prophet - and an empirical evaluation of the models on a range of commodities; (ii) rigorous statistical testing that establishes the ranking of performance and confirms the significance of methodological differences at $p = 0.012$; (iii) the inclusion of 13 external factors, including economic, weather, and policy data, which have delivered significant improvements in accuracy over single variable models; (iv) launch of an intuitive web-based platform that makes advanced forecasting tools accessible, to farmers, traders and policymakers; commodity-specific modelling guidance derived from empirical performance assessments; and a statistical validation framework that supports reliable comparison and selection of models.

The future will be characterized by real-time data assimilation regular forecasts that allow finer market responses, integration of alternative data, including satellite

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images and weather forecasts, combined approaches that integrate different complementary methods and sophisticated deep learning models that enhance the detection of non-linear trends. The expansion of the applications to commodities along with the involvement of additional regions and markets will facilitate the wide usage of agriculture and impact.

REFERENCES

[1] G. E. P. Box and G. M. Jenkins, "Time Series Analysis: Forecasting and Control," Holden-Day, San Francisco, 1970.

[2] S. Kaul and R. Mathur "A review of agricultural commodity price prediction via machine learning " Computers and Electronics, in Agriculture vol. 184 Pp. 106-117, May 2021.

[3] A. K. Mishra and R. P. Singh, "Deep learning techniques for agricultural commodity price prediction: A comprehensive survey," Journal of Agricultural Systems, vol. 178, pp. 102- 119, Sept. 2020.

[4] R. F. Engle and C. W. J. Granger, "Co-integration and error correction: representation, estimation, and testing," Econometrica, vol. 55, no. 2, pp. 251-276, March 1987.

[5] P. S. H. Leelaruban, G. R. Padmanabhan, and K. R. Oduya, "Integrating seasonal and non-seasonal factors for improved agricultural commodity price forecasting," Agricultural Economics, vol. 48, no. 4, pp. 445-456, July 2017.

[6] S. J. Taylor and B. Letham, "Forecasting at scale," The American Statistician, vol. 72, no. 1, pp. 37-45, 2018.

[7] J. Zhang, Y. Li, and W. Chen, "Comparative analysis of time series models for agricultural commodity price prediction," Agricultural Systems, vol. 194, pp. 103-115, October 2021.

[8] R. J. Hyndman and Y. Khandakar, "Automatic time series forecasting: the forecast package for R," Journal of Statistical Software, vol. 27, no. 3, pp. 1-22, July 2008.

[9] M. Kumar and M. Anand, "An application of time series ARIMA forecasting model for predicting agricultural commodity prices in India," Studies in Business and Economics, vol. 9, no. 1, pp. 81-94, April 2014.

[10] D. C. Montgomery, C. L. Jennings, and M. Kulahci, "Introduction to Time Series Analysis and Forecasting," John Wiley & Sons, New York, 2015.

[11] A. S. Tey and M. Brindal "Determinants affecting the uptake of precision farming technologies: an overview for policymakers " International Journal of Agricultural Management vol. 1 No. 4 Pp. 15-25 2012.

[12] "Agricultural Statistics, at a Glance 2024 " Ministry of Agriculture and Farmers Welfare, Government of India New Delhi, India, Tech. Rep., 2024.

[13] Y. Lecun, Y. Bengio, and G. Hinton, "Deep learning," Nature, vol. 521, no. 7553, pp. 436-444, May 2015.

[14] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," International Conference on Learning Representations, 2015.

[15] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," Communications of the ACM, vol. 60, no. 6, pp. 84-90, June 2017.

[16] X. Glorot and Y. Bengio, "Understanding the difficulty of training deep feedforward neural networks," in Proc. 13th International Conference on Artificial Intelligence and Statistics (AISTATS), 2010, pp. 249-256.

[17] S. Ioffe and C. Szegedy, "Batch normalization: accelerating deep network training by reducing internal covariate shift," in Proc. 32nd International Conference on Machine Learning (ICML), 2015, pp. 448-456.

[18] G. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Improving neural networks by preventing coadaptation of feature detectors, " arXiv preprint arXiv:1207.0580, 2012.

[19] R. Salakhutdinov, A. Mnih, and G. Hinton, "Restricted Boltzmann machines for collaborative filtering," in Proc. 24th International Conference on Machine Learning (ICML), 2007, pp. 791-798.

[20] Y. Bengio, P. Lamblin, D. Popovici and H. Larochelle, "Greedy layer-wise training of deep networks, " in Advances in Neural Information Processing Systems (NIPS), 2006, vol. 19, pp.153-160.

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